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Event Relation Extraction Based on Heterogeneous Graph Attention Networks and Event Ontology Direction Induction

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Abstract: Event relation extraction plays a crucial role in constructing an event knowledge graph. However, current models only extract trigger words as event ontology representations, and do not consider node type during information aggregation, resulting in low accuracy in event relation extraction. To address these challenges, we propose an event relation extraction model based on heterogeneous graph attention networks and event ontology direction induction. To enhance the completeness of event information, we incorporate argument role information, in addition to trigger words, into the input text. A novel heterogeneous graph attention framework is proposed to reasonably allocate weights to trigger words, argument roles, and text information, and then perform two levels of aggregation, node-level and semantic-level, in sequence. To improve the accuracy of event direction discrimination, we construct an event ontology subgraph that includes trigger words and arguments to aggregate complete event structure information during direction induction. Finally, we evaluate our model on three datasets, TimeBank-Dense, MATRES, and HiEve, and demonstrate that our model outperforms state-of-the-art models by 1.2%, 0.5%, and 0.8%, respectively, in terms of the Micro-F1 score. Our proposed model provides a promising solution for event relation extraction and can be applied in various natural language processing applications.

Key words: event relation extraction; argument role; heterogeneous graph networks; Event Ontology Direction Induction (EODI)

1 Introduction

Events serve as a significant representation of text-based information and have gained increasing prominence in Natural Language Processing (NLP) in recent years, owing to the existence of various event-based theoretical models and application techniques. Among the different semantic relations that exist between events, the event-time relation and sub-event relation are two typical relations that describe the fundamental structure of an event. Therefore, tasks such as event Temporal Relation Extraction (TRE) and Sub-event Relation Extraction (SRE) are crucial for various downstream applications, including event knowledge graph construction[1, 2], event inference retrospective causation[3, 4], timeline construction[5], and reading comprehension[6]. Events are typically represented by event triggers, which are single or sequential verbs within an event. An event argument refers to an entity mentioned in the text, involved in an

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event, and playing a specific role in the event. The relation between the argument and the event is typically referred to as the argument role. For instance, the event sentences (es_1 and es_2) describe the "deploy" (e_1) and "join" (e_2) events, respectively, and the temporal relation between them can be derived as "after" by utilizing the same argument (250 000 U.S. troops and 250 000 U.S. soldiers), as shown in Fig. 1.

- Event sentence 1 (es_1): U.S. Department of State deploy quarter million U.S. Army soldiers in the Fort Hood, Texas.
- Event sentence 2 (es_2): The army's entire first Calvary division based at Fort Hood, Texas, would join the quarter million U.S. forces already in the region.

Early research on event relation extraction relied on pattern matching and hand-designed syntactic and semantic features[7, 8]. Later, machine learning techniques were utilized to extract lexical and syntactic features for relation extraction. Currently, neural networks and pre-trained language models, such as CNNs[9], RNNs[10], GNNs[10, 11], and Bidirectional Encoder Representations from Transformers (BERT) models[12], are widely employed for relation extraction, enabling tasks such as text categorization. These models have been applied by researchers to identify event temporal and sub-event relation, leading to better

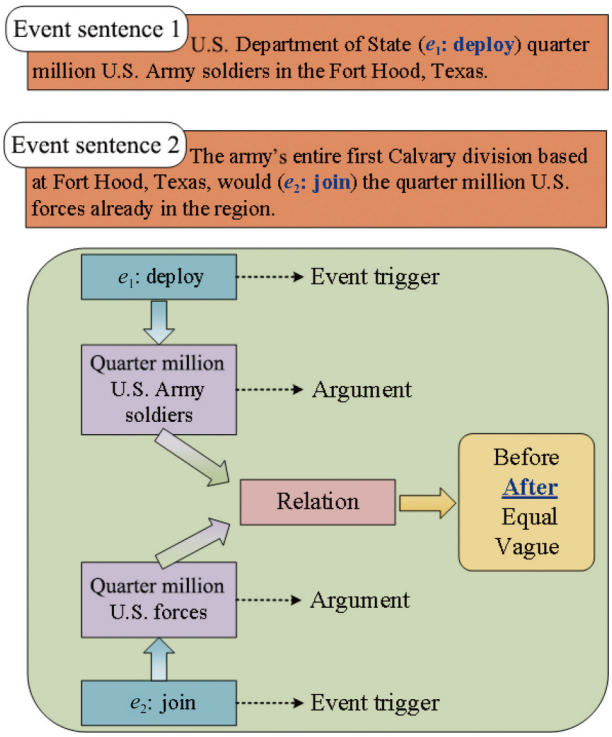

Fig. 1 Relation extraction based on argument information.

accuracy. However, existing models only extract trigger words as event ontology representations, and node type is not considered during information aggregation, resulting in low accuracy in event relation extraction.

To address the challenges of event relation extraction, we propose a model called Heterogeneous Graph Attention Networks and Event Ontology Direction Induction (HGAN-EODI). Firstly, the BERT model is utilized to encode the input text with trigger words and argument roles, producing semantic feature vectors in the input encoder. We then construct a semantic graph in conjunction with the syntactic dependency tree and use the heterogeneous graph networks module to extract the syntactic and semantic information of the text.

To aggregate the information of argument nodes and text nodes in heterogeneous graphs, we propose the HGAN, which incorporates argument role. This mechanism is implemented through heterogeneous graph networks, which can aggregate syntactic-level event information through node-level attention. Semantic features can be aggregated for event relation prediction through semantic-level attention. Node-level attention is used to learn event information between argument or text nodes under the same meta-path and trigger word nodes, while semantic-level attention is able to learn event information between trigger word nodes and different meta-paths. By learning attention coefficients from both node-level and semantic-level attention, event information can be aggregated from different argument and textual semantics.

Furthermore, the event ontology direction induction module is employed to obtain a relational sequence representation based on the event context, which enhances the forward and backward event relation of event pairs. Specifically, the event ontology subgraph is constructed to aggregate trigger words and argument features, generating feature embedding of the event ontology. The feature embedding of the event ontology is then fed into the feed-forward neural network, where feed-forward neural network is used to obtain a forward event pair representation and a backward event pair representation, respectively. The directional sequence representation of the event relation is automatically inferred by continuously learning the weights.

Finally, we feed the event features output from the two modules into the relation extraction classifier to

predict the type of relation for the event pairs. Our proposed HGAN-EODI model achieves state-of-the-art performance on several benchmark datasets, demonstrating its effectiveness in event relation extraction.

In summary, the main contributions of this paper can be summarized as follows:

- To enhance the completeness of event information, we have included argument role information, in addition to trigger words, in the input text.
- Novel node-level aggregation and novel semantic-level aggregation methods are proposed to reasonably allocate the weight of aggregated information by combining trigger words and argument role types in heterogeneous graph attention networks.
- To improve the accuracy of event direction discrimination, we construct an event ontology subgraph to aggregate complete event structure information during direction induction. To achieve this, we utilize Graph Attention Networks Incorporating Argument Role (GAT-IAR) to improve the ability of obtaining event structure information.
- By combining the above methods, we propose the event relation extraction model based on HGAN-EODI. We conduct experiments on the TimeBank-Dense, MATRES, and HiEve datasets and demonstrate that our model outperforms state-of-the-art models in event relation extraction.

The remainder of this paper is organized as follows. Section 2 provides an overview of related work. Section 3 describes the proposed event relation extraction model based on HGAN-EODI. The experimental results are presented in Section 4. Finally, Section 5 concludes the paper and discusses potential avenues for future research.

2 Related Work

This section provides an overview of the current state of research on heterogeneous graph neural networks and event relation extraction.

2.1 Heterogeneous graph neural networks

The direction of research on graph neural networks can be divided into two categories: spectral and non-spectral methods. Spectral methods use a spectral representation of the graph. Bruna et al.[13] generalized the convolution to general graphs by finding the corresponding Fourier basis, while Defferrard et al.[14] used smooth filters in the spectral domain using k-th order Chebyshev polynomials. Wu et al.[15] learned node representations using local first-order approximations of the spectral graph convolution. Niepert et al.[16] normalized each node and its neighbors as the acceptance domain for the convolution operation.

On the other hand, the non-spectral approach defines the convolution directly on the graph and operates on the spatial nearest neighbor group. Chen et al.[17] proposed a framework to generate representations by sampling and aggregating features from the local neighborhood of a node, while Niepert et al.[16] extracted and normalized each node and its neighborhood and used the normalized neighborhood as the receptive domain for the convolution operation.

Meanwhile, attention mechanisms, such as self-attention[18] and soft attention[19], have become one of the most influential mechanisms in deep learning. Several previous works have introduced attention mechanisms for graph-based applications, such as recommendation systems[20, 21]. Inspired by attention mechanisms, Graph Attention neTworks (GATs)[22] were proposed to learn the importance between a node and its neighboring nodes and fuse neighboring nodes for node classification. Wang et al.[23] proposed a heterogeneous graph neural network based on hierarchical attention, including node-level attention and semantic-level attention.

2.2 Relation extraction model

Currently, in the task of event relation extraction, most studies focus on extracting causal and temporal relations. With the development of deep learning, many models have been used in this task. Rink et al.[7] used graphical patterns as features to train a Support Vector Machine (SVM) classifier. Specifically, they constructed a graph representation for a sentence, encoding lexical, syntactic, and semantic information. They automatically extracted graphical patterns from such a graph representation and ranked them according to the relevance of identifying causal relation between events.

Hu et al.[12] adaptively clustered temporal relation features using a pre-trained language model, while Jiang and He[8] proposed a model that incorporates syntactic tree structures in classification. Different ideas for dealing with document-level event causality have also been proposed, and Man et al.[24] designed a reinforcement learning mechanism to select key

contexts from documents.

Since rich linguistic knowledge is contained in syntactic dependencies, a number of approaches based on syntactic dependencies have been proposed. Zhang et al.[25] proposed the use of graph converters to capture temporal knowledge in syntactic graphs. Mathur et al.[26] and Tran et al.[27] used syntactic dependencies along with other knowledge, such as rhetoric, discourse, and semantics, to enrich event representations through node-to-node interactions in different graphs.

Zhuang et al.[11] proposed a new type-guided attention graph convolutional network for event-relation extraction. They constructed an event-specific syntactic dependency graph from the input text, from which local and global dependency knowledge related to events can be derived. Then, a dependency-guided attentional graph convolutional network was designed to learn event representations, where local and global dependency information was utilized to efficiently aggregate semantic contexts between texts. Finally, the task of event relation extraction was accomplished by predicting event relations based on the representations of event pairs and the whole text.

Xu et al.[10] proposed an event information integration model that utilizes multilayer Bi-directional Long- and Short-Term Memory (Bi-LSTM) and attentional mechanisms to integrate event information. Additionally, the model includes a new Relational GAT (RGAT) that incorporates edge attributes. Zhuang et al.[9] infused potential knowledge from the event ontology into the cue text and further introduced a two-layer attention fusion mechanism to enhance the injection of critical knowledge to mitigate knowledge noise.

However, existing models only extract event trigger words as event ontology representations and do not consider node types when aggregating information, leading to low accuracy in event relation extraction. In events, event trigger words typically refer to the core words that describe the event, while event arguments refer to entities mentioned in the text that participate in the event and play specific roles, such as subjects, objects, attributives, etc. In natural language processing, argument roles typically refer to the relations between elements, such as arguments, conclusions, and premises in a text. By identifying and analyzing these argumentative roles, the logical structure and semantic relations in the text can be better understood, resulting in more accurate event relation extraction.

3 Methodology

In this section, we give an overview of HGAN-EODI and the proposed model is demonstrated in detail.

3.1 Model overview

Event relation extraction is the process of extracting relations between events from texts and linking events to create an Event Knowledge Graph (EKG). Given a text $\mathcal{S} = \{\omega_0, \omega_1, \ldots, \omega_n\}$ of n words, where a pair of events are specified as (e_1, e_2), their relation $r(e_1, e_2)$ is identified by event relation extraction model. This paper focuses on the extraction of two types of relation, namely temporal and sub-event relations of events, and the task is typically formalized as text classification given event pairs and contexts.

HGAN-EODI aims to construct a semantic graph with a heterogeneous graph structure and improve the accuracy of event information features by adding labeled argument role information in the text to the process of feature learning and information aggregation of the semantic graph. This enables the model to obtain the degree of correlation between different arguments and the event ontology. The overall structure of the model is shown in Fig. 2, where Z_i and Z_j denote the feature vectors of node i and node j, respeclively, h_{seq} is sequential representation of relation of event pairs and h_{hgan} is the event pair feature vector output by the heterogeneous graph attention network module. The model is consisted of four main modules, including semantic graph construction, heterogeneous graph networks module, event ontology direction induction module, and relation extraction classifier.

3.2 Semantic graph construction

Firstly, the input text data are preprocessed with trigger word label and argument role label, and then processed by the BERT model to generate corresponding feature embedding. The dependency trees obtained from Stanford CoreNLP are utilized to construct the semantic graph.

The BERT model is trained using a large-scale unlabeled corpus, and the resulting text vectors contain rich semantic information. Therefore, this model generates the input text sentences $\{\omega_0, \omega_1, \ldots, \omega_t\}$ into a feature vector matrix V, whose expression is as follows:

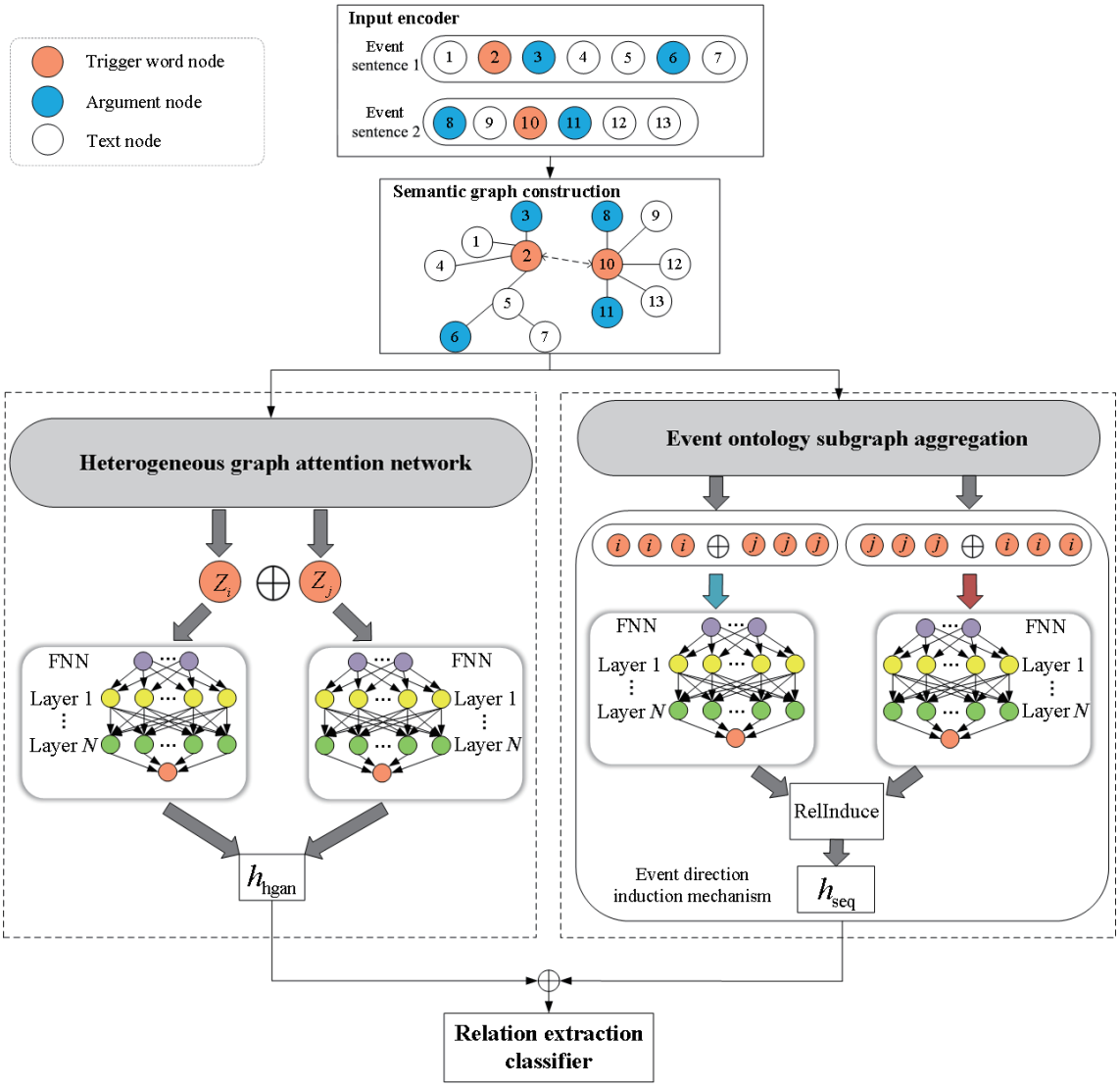

Fig. 2 Overall architecture of our networks (FNN denotes Feedforward Neural Network).

$$V = \begin{pmatrix} \mathrm{cls}_1 & v_{1\times1} & v_{1\times2} & \cdots & v_{1\times768} & \mathrm{sep}_1 \\ \mathrm{cls}_2 & v_{2\times1} & v_{2\times2} & \cdots & v_{2\times768} & \mathrm{sep}_2 \\ \mathrm{cls}_3 & v_{3\times1} & v_{3\times2} & \cdots & v_{3\times768} & \mathrm{sep}_3 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ \mathrm{cls}_t & v_{t\times1} & v_{t\times2} & \cdots & v_{t\times768} & \mathrm{sep}_t \end{pmatrix} \tag{1}$$

where t is the sequence length, 768 is the dimension of the word vector. For element $v_{m\times n}$, m represents the m-th word, and n represents the dimension of the word vector. Each sentence of the input can be converted into a feature vector matrix by the Bert model. The matrix V is constructed by encoding sentences with t words using BERT, where v_m represents the word vector corresponding to the i-th word. The special markers "cls" and "sep" denote the beginning and end of the word, respectively. The 748-dimensional word vector v lies between these two markers.

Given two event sentences $[\mathrm{es}_1; \mathrm{es}_2]$, utterance dependency parsing using Stanford CoreNLP yields two dependency trees for the two event sentences. These dependency trees are then used as inputs to construct the semantic graph.

To represent an input sentence in a semantic graph structure, the syntactic structure of the sentence is constructed in a syntactic dependency tree. In this tree, words are represented as nodes, and dependencies between words are represented as edges. This allows for the conversion of a sentence into a semantic graph structure. The nodes in the semantic graph are divided into two categories: event ontology nodes and text nodes. The event ontology nodes are further divided into two categories: trigger word nodes and argument nodes. The text nodes represent non-triggering words.

This categorization of nodes facilitates the extraction of syntactic and semantic information from the input text, which is then utilized in the HGAN-EODI model for event relation extraction.

The process of constructing a semantic graph from a syntactic dependency tree is illustrated in Fig. 3. It begins from an event trigger word and recursively searching for p-hop nodes in each dependency tree to form two subgraphs, es_1 and es_2. An edge of trigger-trigger type is then used to connect the two event triggers, generating a complete semantic graph g.

To enhance information fusion, four types of edges are set up in g, namely:

- **Edges of primitive type:** this type of edges are primitive dependency edges generated by syntactic dependency parsing.
- **Edges of the reverse type:** this type of edges are additional edges as opposed to edges of the original type.
- **Edges of type trigger-trigger:** this type of edge is an additional edge that connects between event triggers.
- **Self-pointing type of edges:** this type of edges are additional edges that point to the node itself.

This construction of the semantic graph allows for the aggregation of syntactic and semantic information from the input text, which is then used in the heterogeneous graph networks module and event ontology direction induction module.

3.3 Heterogeneous graph networks module

Incorporating trigger word and argument role by HGAN is utilized to extract the syntactic and semantic information of the input text and aggregate the information of trigger, argument, and text nodes in the heterogeneous graphs, as shown in Fig. 4.

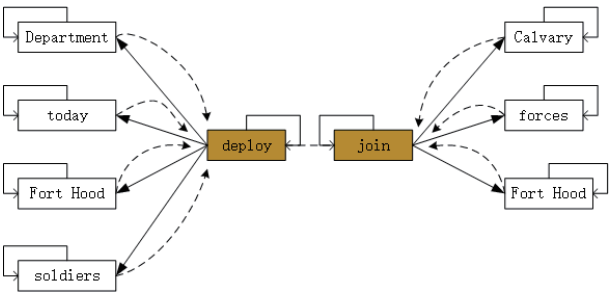

Fig. 3 Partial semantic graph of event sentence 1 (U.S. Department of State deploy quarter million U.S. Army soldiers in the Fort Hood, Texas) and event sentence 2 (The army's entire first Calvary division based at Fort Hood, Texas, would join the quarter million U.S. forces already in the region).

Specifically, HGANs incorporating trigger word and argument role can aggregate syntactic-level event information through node-level attention and semantic features for event relation prediction through semantic-level attention. Node-level attention aims to learn event information between argument or text nodes under the same meta-path and trigger word nodes, while semantic-level attention is able to learn event information between trigger word nodes and different meta-paths.

3.3.1 Node-level aggregration of capture event structure

The features of nodes under the same path are first aggregated through the node-level attention mechanism. The feature vector ω_i corresponding to $\{\omega_0, \omega_1, \ldots, \omega_t\}$ is first initialised using h_i, which is calculated as follows:

$$h_i = \omega_i \tag{2}$$

where h_i is word vector of node i.

Then, self-attention is used to learn the weights between various nodes. Given a node pair (i, j) connected via path Φ, node-level attention e_{ij}^{Φ} can learn the importance between node pairs, which implies the importance of node j to node i. We propose a novel node-level aggregation method that leverages both the feature embedding of node pairs and the representation of text type, argument role, and trigger word type, to calculate the weight score of aggregated nodes. The path-based importance e_{ij}^{Φ} of node pair (i, j) can be expressed as follows:

$$e_{ij}^{\Phi} = \text{LeakyReLU}\,(a\,(W_r\,[h_i, h_j] \odot W_{\text{dep}}\,[d_i, d_j])) \tag{3}$$

where a is a learnable weight vector; LeakyReLU $(\cdot)$ is a nonlinear activation function with a negative input slope of 0.2; W_{dep} is a shared learnable linear transformation weight matrix; W_r is a shared learnable linear transformation weight matrix for relation r; d denotes the representation of text type, i.e., argument role or trigger word type, and "$\odot$" denotes the concatenation operation.

Subsequently, the softmax function is used to normalise the neighbouring nodes under the same path, which is calculated as follows:

$$\alpha_{ij}^{\Phi} = \text{softmax}\,(\text{e}_{ij}^{\Phi}) = \frac{\exp\,(\text{e}_{ij}^{\Phi})}{\sum_{k \in N_i^{\Phi}} \exp\,(\text{e}_{ik}^{\Phi})} \tag{4}$$

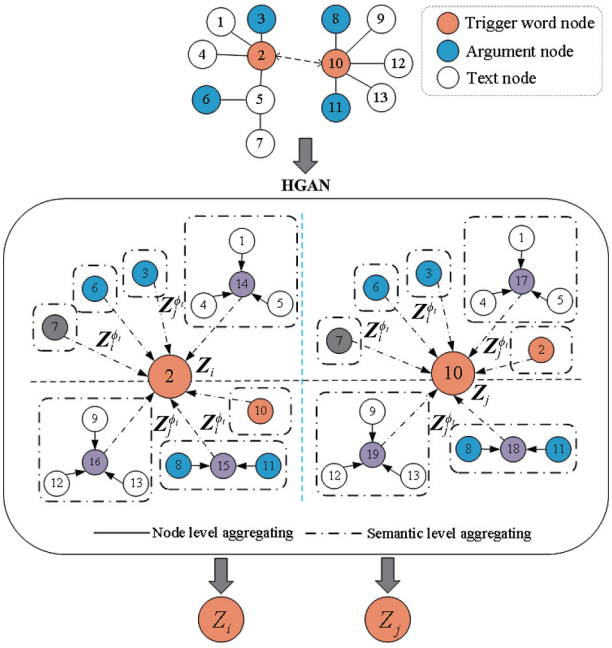

Fig. 4 Overall framework of the proposed HGANs incorporating text type, trigger word type, and argument role.

where N_i^{Φ} denotes the neighbour node of node i based on path Φ; $\alpha_{i,j}^{\Phi}$ is the attention weight of node j to node i. After that, the path-based feature embedding of node i can be aggregated by the feature vectors of its neighbours, and its corresponding calculation formula is as follows:

$$z_i^{\Phi} = \sigma\left(\sum_{j \in N_i^{\Phi}} \alpha_{i,j}^{\Phi} h_i\right) \tag{5}$$

where $\sigma(\cdot)$ is a nonlinear activation function; z_i^{Φ} is the feature vector of node i aggregated by neighbouring nodes based on the path Φ.

3.3.2 Semantic-level aggregration of capture semantic structure

Given the set of paths $\{\Phi_0, \Phi_1, \ldots, \Phi_P\}$, after applying node-level attention to the node feature vectors, a set of P semantically specific node feature vectors can be obtained, denoted as $(Z_{\Phi_0}, Z_{\Phi_1}, \ldots, Z_{\Phi_P})$.

The semantic-level attention mechanism in the HGANs incorporating trigger word and argument role processes the node feature vector sets of node i. Firstly, the attention value of each path is calculated, which applies not only to a single node, but to all nodes.

In the case of a given path type Φ_i, we extract the representation vectors of all nodes that are semantically specific, as well as the feature representations of all node types that exist within the path. Afterward, a nonlinear transformation, such as a layer of Multi-Lager Perceptron (MLP), is used to transform representation vectors of all nodes that are semantically specific. The average of the all node types representation vector is then used as the importance of the semantic-level aggregation vector transformed embedding. Furthermore, we determine the average value of all semantic-specific node embeddings, which can be interpreted as the attention value of each path type. The specific calculation method is as follows:

$$\omega_{\Phi_i} = \frac{1}{V}\sum_{i \in V}\left(\frac{1}{N}\sum_{j \in N^{\Phi}} \tanh\left(W_{\Phi} \cdot D_j + b_1\right)\right)^{\mathrm{T}} \times \tanh\left(W_z \cdot z_i^{\Phi} + b_2\right) \tag{6}$$

where W_{Φ} and W_z are weight matrixes, b_1 and b_2 are the bias vectors, D_j is the mark of text type, trigger word type, and argument rule of the node, which is in the path Φ_i, V is the set of nodes in the semantic graph, N^{Φ} is the set of nodes in the path Φ_i, and ω_{Φ_i} denotes the attention value of path Φ_i.

Attention weights are obtained by normalising ω_{Φ_i} by softmax function. The attention weight of path Φ_i, denoted as β_{Φ_i}, can be obtained by normalising the above attention values of all paths using the softmax function as follows:

$$\beta_{\Phi_i} = \frac{\exp\left(w_{\Phi_i}\right)}{\sum_{i=1}^{P} \exp\left(w_{\Phi_i}\right)} \tag{7}$$

where P denotes the number of path types connected to node i. For different nodes, there exists β_{Φ_i} different situation. After that, these semantic feature vectors can be fused to obtain the final feature vector Z_i as follows:

$$Z_i = \sum_{i=1}^{P} \beta_{\Phi_i} \cdot Z_{\Phi_i} \tag{8}$$

Subsequently, the feature of the output trigger word nodes i and j are fed into the feed-forward neural network. Specifically, the feed-forward neural network is used to fuse the feature vectors represented by the event pairs (i, j) output from the path graph attention mechanism into the representations of the event pairs. The calculation method is as follows:

$$h_{\text{hgan}} = (\mathrm{FFN}_1\left(Z_i \odot Z_j\right))W_1\left(\mathrm{FFN}_2\left(Z_j \odot Z_i\right)\right) \tag{9}$$

where W_1 denotes the weight matrix, and FFN_1 and

FFN_2 are two FFNs.

3.4 EODI module

The EODI module enhances the forward and backward event relation of event pairs by constructing an event ontology subgraph to aggregate trigger words and argument features. The aggregation of the event ontology subgraph is shown in Fig. 5. The feature of the event ontology is fed into a feed-forward neural network to obtain a forward event pair representation and a backward event pair representation. The directional sequence representation of the event relation is automatically inferred by continuously learning the weights.

The feature vector matrix V generated by the pre-trained BERT model is input to the EODI mechanism. Event ontology subgraph aggregation is utilized to obtain the event-based representation of the contextual sequence of the relation. Given two event sentences $[es_1; es_2]$ constituting an event pair and their semantic graph G, all the event ontology nodes in it are extracted to construct the event ontology subgraph before GAT-IAR. GAT-IAR is proposed for better aggregation of event ontology information. The specific process is as follows.

For each node i in the subgraph, the attention coefficient between it and each neighboring node j at layer l is

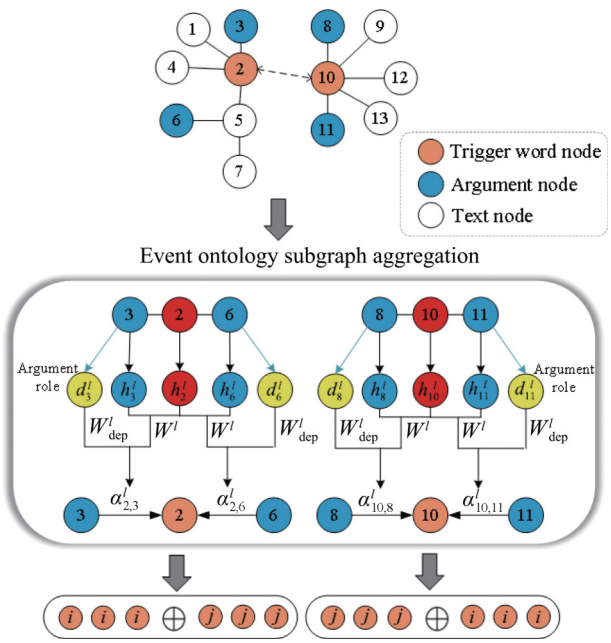

Fig. 5 Event ontology subgraph aggregation.

$$u_{i,j}^{l} = \text{LeakyReLU}\left(a\left(W^{l}[h_i^{l}, h_j^{l}] \odot W_{\text{dep}}^{l} d_j^{l}\right)\right) \tag{10}$$

where h_i^l and h_j^l are feature vectors of node i and node j; d_j^l is text markers, trigger word markers or argument role marker of the node; W_{dep}^l and W^l are the shared learnable linear transformation weight matrixes of dependency. Softmax function is used for the normalisation of the attention coefficients,

$$\alpha_{i,j}^{l} = \text{softmax}\left(u_{i,j}^{l}\right) = \frac{\exp\left(u_{i,j}^{l}\right)}{\sum_{k \in N} \exp\left(u_{i,k}^{l}\right)} \tag{11}$$

where N is a set of neighbouring nodes of node i. Thus, we get the feature vector of node i at layer $l+1$ using the following equation:

$$h_i^{l+1} = \sigma\left(\sum_{j \in N} \alpha_{i,j}^{l}\left(W^{l} h_j^{l} \odot W_{\text{dep}}^{l} d_j^{l}\right)\right) \tag{12}$$

The EODI obtains a forward event pair representation u_{forward} and a backward event pair representation u_{backward} using a layer of feed-forward neural networks. By considering both forward and backward perspectives, the model can further mine context-based event relation information. The directional sequence representation of the event relation is automatically inferred by continuously learning the weights. The computational formula is as follows:

$$u_{\text{forward}} = \text{FFN}_3\left(h_i \oplus h_j\right) \tag{13}$$

$$u_{\text{backward}} = \text{FFN}_4\left(h_j \oplus h_i\right) \tag{14}$$

$$h_{\text{seq}} = u_{\text{forward}} W_2 u_{\text{backward}} + W_3\left(u_{\text{forward}} \oplus u_{\text{backward}}\right) + b \tag{15}$$

where FFN_3 and FFN_4 are two FFNs, W_2 and W_3 are the trainable parameters, and b is the bias.

3.5 Relation extraction classifier

To fully utilize the event sequence representation and event ontology representation obtained by the FNN of the EODI module, the feature vectors output from this module are connected to the feature vectors of aggregated text and event information output from the Heterogeneous graph networks module. A gate mechanism is utilized to dynamically fuse the two different types of information.

The output vectors are then connected to the sequence representations output by the FNN using the gating mechanism. The gating mechanism formula is

$$g_0 = \text{sigmoid}\left(W_4 \cdot h_{\text{seq}} + W_5 \cdot h_{\text{hgan}} + b_f\right) \tag{16}$$

$$h_{(i,\,j)} = g_0 \odot h_{\text{seq}} + (1 - g_0) \odot h_{\text{hgan}} \tag{17}$$

where W_4 and W_5 denote the trainable parameter matrixes, and b_f is the deviation scalar.

Relation extraction classifier is composed using a deep FNN, which connects the final event pairs representation $h_{(i,\,j)}$ and the event nodes h_i and h_j and non-event nodes v_i and v_j of the semantic graph inputs, as well as the feature vectors v_{hgan} of the non-event nodes that have passed through the HGAN, in order to predict the event relation between events t_i and t_j, with the following equations:

$$p\,(\text{re}|t_i, t_j) = \text{softmax}\,(\text{FNN}([h_i;\, h_j;\, v_i;\, v_j;\, v_{\text{hgan}};\, h_{(i,\,j)}])) \tag{18}$$

where "re" is the relation of event, and $p\,(\text{re}|t_i,\, t_j)$ denotes probability of an event pair. In order to obtain a high performance model, the model is optimised using the prediction results and the loss function,

$$J(\theta) = -\sum_{i=1}^{N_E} (\log p\,(\text{re}_{t_i,\,t_j}|t_i,\, t_j)) + \frac{\lambda}{2}\|\Theta\|^2 \tag{19}$$

where N_E denotes the number of event sentence pairs; $\text{re}_{t_i,\,t_j}$ denotes the relation between event pair $(t_i,\, t_j)$; Θ is the set of model parameters; and λ denotes the parameters of L2 regularisation.

4 Experiment

4.1 Datasets

The event temporal relation extraction datasets MATRES, TB-Dense, and the subevent relation extraction dataset HiEve are used to verify the proposed HGAN-EODI model.

- **TB-Dense**[28] makes a more fine-grained division of temporal relation between events, which includes six relation types: BEFORE (B), AFTER (A), SIMULTANEOUS (S), INCLUDES (I), IS INCLUDED (II), and VAGUE (V).
- **MATRES**[29] is the latest and most widely used bench-mark dataset for TRE. It mainly annotates four relations: Before, After, Equal, and Vague.
- **HiEve**[30] is derived from the news corpus, which provides detailed annotations on the four subevent relations (i.e., SubSuper (an event is a subevent of another event), SuperSub (an event contains another event), Coref (coreference event), and NoRel (indeterminate event relation)) between events.

The detailed overall data distributions of the TB-Dense, MATRES, and HiEve dataset are shown in Table 1.

4.2 Baselines

We selected a series of representative models as the baselines for comparison. HGAN-EODI is compared with the following baselines:

- **KEETE**[31]: A knowledge enhanced method for event and TempRel extraction, which integrates the temporal common-sense knowledge, data augmentation, and focal loss function into one single extraction system.
- **Probabilistic Box**[32]: A box event relation extraction that strengthens the logical consistency between predicted event types by representing each event as a box representation.
- **RA-CasRel**[33]: A relation aware embedding Mechanism for relation extraction is applied to CasRel.
- **SDLG**[34]: A syntax-based dynamic latent graph network for event temporal relation extraction and subevent relation extraction.
- **OntoEnhance**[9]: It injects the latent knowledge in the event ontology into the prompt text to address the issue of knowledge missing. A dual-stack attention fusion mechanism is further introduced to enhance the injection of key knowledge to alleviate knowledge noise.
- **RGAT**[10] **and RGAT-IEA**[10]: They build a semantic dependency graph through dependency parsing, model a semantic graph that considers the edges' attributes by using top-k attention mechanisms to learn hidden semantic contextual representations, and finally predict event temporal relations.

It should be noted that since MATRES is a relatively

Table 1 Data statistics for datasets TB-Dense, MATRES, and HiEve.

Dataset	Training		Development		Test	
	Number of documents	Number of pairs	Number of documents	Number of pairs	Number of documents	Number of pairs
TB-Dense	22	4032	22	4032	9	1427
MATRES	260	10 888	21	1852	20	840
HiEve	80	35 001	–	–	20	7093

new dataset, it is difficult to find baselines that perform well on MATRES, TB-Dense, and HiEve. Therefore,we tend to choose different baseline models for different datasets.

4.3 Parameter settings

Our computational hardware environment consists of an Intel Xeon Glod 5218 CPU, NVIDIA GeForce RTX 2080Ti GPU, and 64 GB RAM. The software environment is implemented using Pycharm community 2019.2 and Pytorch 1.8.0 framework. The JMEE[35] model is used to extract trigger words and argument element tokens. We select the "BERT-large-uncased" version of the pre-trained BERT to initialize the token embedding vector. The dimension of the token embedding vector is set to 768. The dimensions of the part-of-speech vector and the dependency vector are both set to 200 and are initialized randomly. The dimensions of the hidden representative vector are 256. We randomly initialize parameters and optimize the model with Adam[36]. We set the learning rate to 0.005, the regularization parameter to 0.001, the dimension of the semantic-level attention vector q to 128, and the dropout of attention to 0.6. The layer of the event ontology graph attention mechanism is set to 2 layers.

4.4 Results and analysis

We use Micro-F1 score as an assessment of model performance. Its related metrics are defined as follows:

$$P = \frac{N_C}{N_P},$$
$$R = \frac{N_C}{N_I},$$
$$\text{Micro-F1} = \frac{2PR}{P+R} \tag{20}$$

where N_C, N_P, and N_I are the number of positive classes that correctly predict the relation between events, the number of all predicted outcomes that are positive classes, and the number of all pairs of events in the dataset for which the relation is a positive class, respectively.

Table 2 presents the results on the TB-Dense dataset. The experimental results demonstrate that the overall performance of our proposed model outperforms several baseline models, with an overall relation extraction performance improvement of 1.2% compared to the RGAT-IEA model. For each specific event relation, the performances of the event relations AFTER, BEFORE, and IS_INCLUDED are increased by 1.3%, 0.9%, and 1.1%, respectively. Consistent with the previous baseline model, the training samples of relation SIMULTANEOUS are too few to predict. The performances of relations INCLUDES and VAGUE are reduced by 0.7% and 0.1%, respectively. The experimental findings indicate that our proposed event ontology subgraphs and heterogeneous graph attention networks, which incorporate trigger words and argument roles, can effectively capture the degree of association between diverse arguments and the event ontology, as well as the forward and backward relation representation of event pair sequences. This approach can improve the performance of event relation extraction.

Table 2 Experimental results on TB-Dense. The numbers in bold and underlined denote the best and second best models, respectively.

(%)

Model	A	B	S	I	II	V	Micro-F1
KEETE[31]	68.4	67.2	–	18.9	33.7	68.9	65.6
RA-CasRel[33]	52.0	50.3	–	12.2	33.1	63.6	57.1
SDLG[34]	–	–	–	–	–	–	68.4
OntoEnhance[9]	69.0	67.8	–	**42.1**	49.7	68.3	66.0
RGAT[10]	71.0	72.6	–	41.2	**52.5**	68.8	69.6
RGAT-IEA[10]	<u>72.1</u>	<u>72.8</u>	–	<u>41.9</u>	51.2	**69.2**	<u>70.7</u>
Ours	**73.4**	**73.7**	–	41.2	<u>52.3</u>	<u>69.1</u>	**72.1**

To verify the generalizability of our proposed model, we conduct experiments on the MATRES and HiEve datasets. The experimental results on these datasets are presented in Tables 3 and 4. Our model demonstrates superior performance when compared to the state-of-the-art models on both datasets, thereby substantiating its efficacy. Despite slightly lower performance than that on the MATRES dataset, our model still outperforms RGAT-IEA by 0.2% and 1.0% on

Table 3 Experimental results on MATRES. The numbers in bold and underlined denote the best and second best models, respectively.

(%)

Model	*P*	*R*	Micro-F1
KEETE[31]	–	–	76.6
Probabilistic Box[32]	–	–	77.1
RA-CasRel[33]	72.6	79.8	76.0
OntoEnhance[9]	78.5	85.0	81.6
RGAT[10]	<u>77.6</u>	85.6	81.4
RGAT-IEA[10]	**78.9**	<u>85.8</u>	<u>82.2</u>
Ours	78.5	**86.7**	**82.4**

Table 4 Experimental results on HiEve. The numbers in bold and underlined denote the best and second best models, respectively.

(%)

Model	*P*	*R*	Micro-F1
Probabilistic Box[32]	–	–	60.6
SDLG[34]	65.5	61.9	63.6
OntoEnhance[9]	58.0	68.2	62.7
RGAT[10]	57.7	68.0	62.4
RGAT-IEA[10]	<u>59.3</u>	<u>68.9</u>	<u>63.7</u>
Ours	**60.3**	**69.7**	**64.7**

MATRES and HiEve datasets, respectively. These results suggest that our approach represents a promising solution for event extraction tasks, and has the potential to be further optimized for even better performance.

In this study, we investigate the impact of varying levels of argument annotation on the accuracy of event relation extraction. Specifically, we employ argument role datasets that are annotated with different percentages (0%, 25%, 50%, 75%, and 100%), and utilize them as input data for our HGAN-EODI model. Our experimental results reveal a positive correlation between the number of argument role annotations and the precision of event relation extraction, as illustrated in Fig. 6. Notably, we observe that the greatest performance improvement is achieved when the number of annotations increased from 25% to 75%. Conversely, the effect of increasing annotations from 75% to 100% was relatively minor.

In order to validate the robustness of our model, we conduct noise experiments by introducing 1%, 2%, and 3% erroneous event relations in the training set. Each configuration is tested with 10 different random seeds. The results of the noise experiments are shown in Fig. 7. The experimental results demonstrate that our proposed model exhibits good robustness in TB-Dense

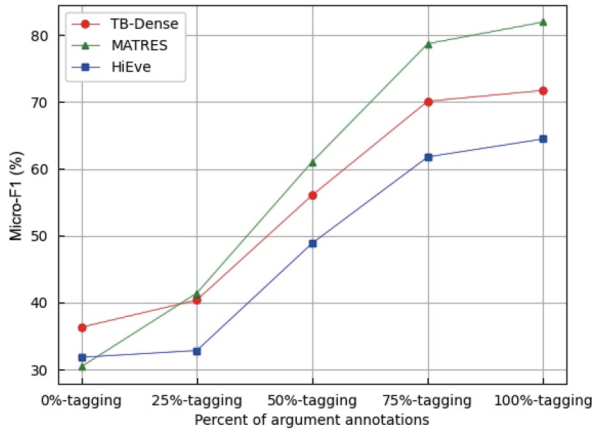

Fig. 6 Experimental results of the percent of argument annotations on the accuracy of event relation extraction.

and MATRES datasets, and the average of Micro-F1 score does not decrease by more than 2% under 3% noise. Although the average of Micro-F1 score of our model on the larger HiEve dataset decreased by 5.8% under 3% noise, it still achieves good stability under 1% and 2% noise. It is noteworthy that our model can maintain good performance under various noise levels, indicating its potential for practical applications.

Furthermore, to investigate the influence of the quantity of GAT layers on the performance of event ontology subgraph aggregation, a series of corresponding experiments are conducted. Same as before, each configuration is tested with 10 different random seeds. The outcomes of these experiments are presented in Fig. 8. The experimental results obtained from three datasets indicate that the performance of event relation extraction is optimized best when the number of GAT layers is 2.

4.5 Ablation studies

In this study, we investigate the effectiveness of two modules, namely, HGAN-ITA and EODI, in enhancing event relation extraction performance. We evaluate the

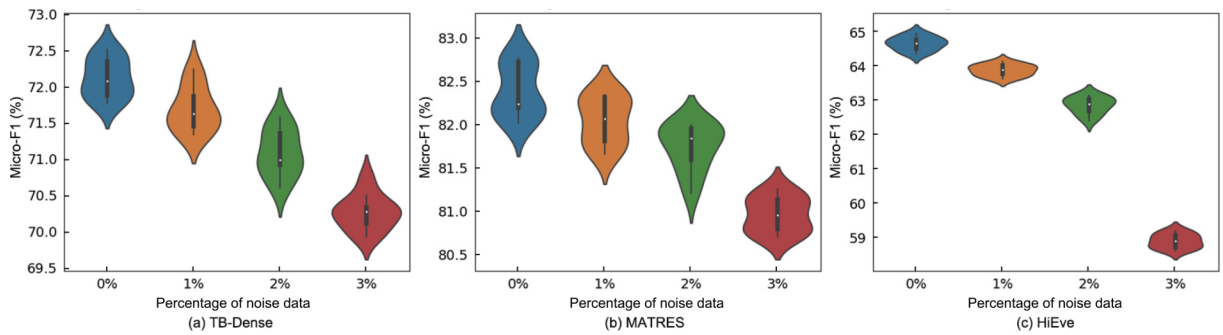

Fig. 7 Noise experimental results on TB-Dense, MATRES, and HiEve datasets.

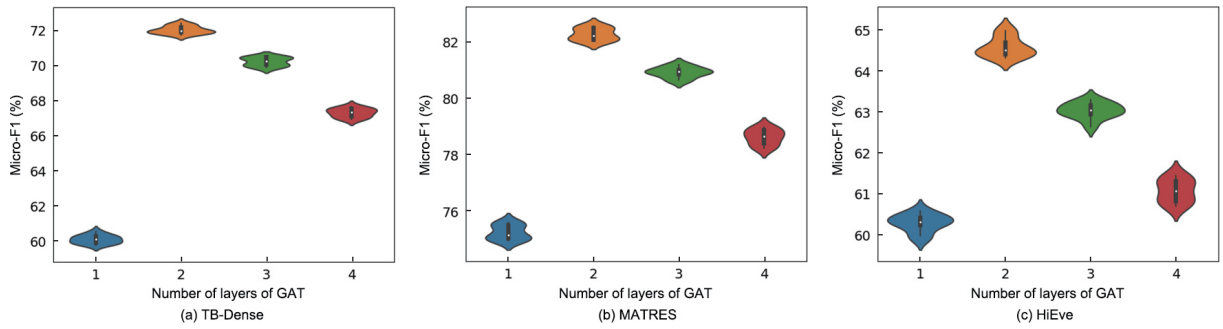

Fig. 8 Model performances of different GAT layers.

performance of these models on the TB-Dense dataset, and the results are presented in Table 5. To compare the proposed models with the baseline model, we use RGAT-IEA (denoted as RI in Table 5) as an alternative. We label it as RI-EODI and evaluate the impact of HGAN-ITA on relational extraction performance. Similarly, we remove the event ontology direction induction component from the model to assess the effect of this component on relation extraction performance, which we denote as HGAN-ITA.

The superiority of HGAN-ITA in event relation extraction performance is evident from the comparison of the Micro-F1 scores of HGAN-ITA and RI, as well as HGAN-EODI and RI-EODI. Likewise, the effectiveness of EODI in improving event relation extraction performance can be observed by comparing the Micro-F1 scores of RI and RI-EODI, as well as HGAN-ITA and HGAN-EODI. Our experimental results clearly demonstrate the effectiveness of both HGAN-ITA and EODI in enhancing event relation extraction performance.

Furthermore, we conduct ablation experiments on the MATRES and HiEve datasets, and the results are presented in Table 6. Although the precision (P) on MATRES and recall (R) on HiEve may not fully demonstrate the superiority of the models, the significant improvement in event relation extraction performance achieved by HGAN-ITA and EODI is evident from other data and Micro-F1 scores.

Table 5 Ablation studies on TB-Dense. The numbers in bold and underlined denote the best and second best models, respectively.

(%)

Model	A	B	S	I	II	V	Micro-F1
RI	72.1	72.8	–	**41.9**	51.2	**69.2**	70.7
HGAN-ITA	<u>73.1</u>	<u>73.6</u>	–	41.0	<u>51.9</u>	<u>69.1</u>	<u>71.4</u>
RI-EODI	72.3	73.1	–	<u>41.5</u>	51.6	68.9	70.9
HGAN-EODI	**73.4**	**73.7**	–	41.2	**52.3**	<u>69.1</u>	**72.1**

Table 6 Ablation studies on MATRES and HiEve, respectively. The numbers in bold and underlined denote the best and second best models, respectively.

(%)

Model	MATRES			HiEve		
	P	R	Micro-F1	P	R	Micro-F1
RI	**78.9**	85.8	82.2	59.3	68.9	63.7
HGAN-ITA	78.4	<u>86.5</u>	<u>82.3</u>	59.7	**69.8**	<u>64.4</u>
RI-EODI	<u>78.7</u>	86.3	<u>82.3</u>	<u>59.8</u>	68.5	63.9
HGAN-EODI	78.5	**86.7**	**82.4**	**60.3**	<u>69.7</u>	**64.7**

5 Conclusion

In contrast to existing event relation extraction methods that solely rely on trigger words, we incorporate argument role information to enhance the completeness of event information. Based on this, we propose a novel event relation extraction model that utilizes HGAN and EODI. Specifically, our model proposes novel node-level and semantic-level aggregation methods to extract event and text information from the semantic graph, while the event ontology subgraph is constructed to aggregate event structure information and conduct direction induction. Experimental results demonstrate that our model outperforms state-of-the-art models in terms of accuracy performance, providing a promising solution for event relation extraction that can be applied in various natural language processing applications.

In the future work, we plan to explore how to extend our model to handle dynamic semantic graph scenarios. This would require the development of efficient and effective methods for handling changes in the semantic graph over time, which is a challenging research problem that has yet to be fully addressed.

In addition, the integration of multimodal data and

event knowledge graph remains an important research direction for us. We believe that this presents exciting opportunities for interdisciplinary collaboration and innovation in the fields of natural language processing, computer vision, knowledge representation, and machine learning.

With the development of large language models, they have been widely applied in natural language processing tasks. Therefore, using large language models for event relation extraction tasks has great potential. Large language models can process large amounts of language data and capture contextual information in language, resulting in more accurate identification of relationships between events. Additionally, large language models can learn the commonalities and characteristics of different types of events, making event relation extraction possible in various fields. However, large language models require a significant amount of training data and processing parameters, which may lead to overfitting and misunderstanding. Therefore, further research on improving the performance of large language models in event relation extraction tasks to enhance their accuracy and reliability is one of our future research directions.

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